

FuelGauge.h Copy-paste code:

```
#ifndef FUELGAUGE_H

class FuelGauge {
private:
    int fuel;
    const int MAXIMUM_CAPACITY = 15;
public:
    FuelGauge (const int f) {
        fuel = f;
    }

    int getGallons() const {
        return fuel;
    }

    void addToTank(const int gal) {
        fuel += gal;
    }

    void removeFromTank(const int gal) {
        fuel -= gal;
    }

    void report() const;
};

void FuelGauge::report() const{
    if (fuel == 0) {
        std::cout << "Tank is empty!" << std::endl;
    }
    else {
        std::cout << "Fuel: " << fuel << std::endl;
    }
}

#endif // FUELGAUGE_H
```

Odometer.h Copy-paste code:

```
#ifndef ODOMETER_H
#define ODOMETER_H

class Odometer {
    friend class FuelGauge;
private:
    int mileage;
    int mileCounter = 0;
public:
    Odometer(const int miles) {
        mileage = miles;
    }

    void advance(int x, FuelGauge &gauge);
    void report() const;
};

void Odometer::advance(const int x, FuelGauge &gauge) {
    mileage += x;
    mileCounter++;
    if (mileCounter >= 24) {
        gauge.removeFromTank(1);
        mileCounter = 0;
    }

    if (mileage > 999999) {
        mileage = 0;
    }
}

void Odometer::report() const {
    std::cout << "Odometer: " << mileage << "\n";
}

#endif // ODOMETER_H
```

main.cpp Copy-paste code:

```
//DO NOT MODIFY THIS SECTION
#include <iostream>
#include "FuelGauge.h"
#include "Odometer.h"
using namespace std;

int main()
{
    Odometer odom( 999990 );    //Create and initialize
    FuelGauge gauge( 0 );       //Create and initialize to zero
    odom.report();              //Display info from odom
    gauge.report();             //Display info from gauge
    gauge.addToTank( 1 );       //Add one gallon
    gauge.report();             //Display new info from gauge
    while( gauge.getGallons()>0 ) //Repeat while we still have fuel
    {
        odom.advance( 1, gauge ); //Advance one mile. Include reference to
gauge
        odom.report();            //Report info from odom
        gauge.report();          //Report info from gauge
    }
    return 0;
}
```

Screenshot of program output:

```
Odometer: 999990      Tank is empty!
Fuel: 1
Odometer: 999991      Fuel: 1
Odometer: 999992      Fuel: 1
Odometer: 999993      Fuel: 1
Odometer: 999994      Fuel: 1
Odometer: 999995      Fuel: 1
Odometer: 999996      Fuel: 1
Odometer: 999997      Fuel: 1
Odometer: 999998      Fuel: 1
Odometer: 999999      Fuel: 1
Odometer: 0           Fuel: 1
Odometer: 1           Fuel: 1
Odometer: 2           Fuel: 1
Odometer: 3           Fuel: 1
Odometer: 4           Fuel: 1
Odometer: 5           Fuel: 1
Odometer: 6           Fuel: 1
Odometer: 7           Fuel: 1
Odometer: 8           Fuel: 1
Odometer: 9           Fuel: 1
Odometer: 10          Fuel: 1
Odometer: 11          Fuel: 1
Odometer: 12          Fuel: 1
Odometer: 13          Fuel: 1
Odometer: 14          Tank is empty!

Process finished with exit code 0
```